# Pengcheng Wang

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## EDUCATION

**University of Illinois, Urbana-Champaign** **Aug 2025 to May 2027**

*Master of Science in Computer Science*

- **Advisor**: Tong Zhang
- **Research Area**: Large Language Model, Post-training (SFT & RL), Agentic Workflow

**University of Toronto,** Toronto, ON **Sept 2020 to May 2025**

*Bachelor of Applied Science in Computer Engineering, Minor in AI Engineering*

- **CGPA**: 3.82/4.0
- **Honors & Awards**: Consecutive four years awarded the Engineering Dean's honor list; Obtained $6400 T-CAIREM 2022 Healthcare AI research scholarship; Honor graduates.

## CONFERENCE & JOURNAL PROCEEDINGS

- Y Wu, X Gao, Z Liu, **P Wang**, Z Wu, Y Li, T Zhang, T Liu, T Liu, X Li, Decoding cortical folding patterns in marmosets using machine learning and large language model. In: NeuroImage, Volume 308, 2025.
- H Zhang*, **P Wang***, S Diao, Y Lin, R Pan, H Dong, D Zhang, P Molchanov, T Zhang, Entropy-Regularized Process Reward Model. In: TMLR, June 2025.
- S Diao, **P Wang,** Y Lin, T Zhang, Active prompting with chain-of-thought for large language models. In: Association for Computational Linguistics *(ACL) Conference 2024*.

## PROJECT & RESEARCH EXPERIENCE

**Chemistry Research Agent** **Aug 2025 to Current**

*Supervisor: Prof. Tong Zhang (UIUC)*

- Building LLM agents with the capability of conducting molecular chemistry research tasks.
- Optimize agentic workflow and context management.

**Entropy-Regularized Process Reward Model** **June 2024 to Oct 2024**

*Supervisors: Dr. Shizhe Diao (Nvidia) & Prof. Tong Zhang (UIUC)*

- Implemented Entropy-Regularized process reward in reward model training and evaluation.
- Implemented RAFT and DPO algorithm in RLHF pipeline to improve policy model's math reasoning ability.
- Project Page: https://hanningzhang.github.io/math-prm/

**Active Prompting with Chain-of-Thought for Large Language Models** **Dec 2022 to March 2023**

*Supervisors: Dr. Shizhe Diao & Prof. Tong Zhang (HKUST)*

- Designed and implemented experiments for prompt question selection strategy through active learning.
- Achieved new SOTA on multiple reasoning benchmarks with the proposed prompting framework using GPT family models.

**ICU RL Benchmark Dataset for Dosage Prediction** **May 2022 to Sept 2022**

*Worked at The Hospital for Sick Children, under the supervision of Dr. Alistair Johnson.*

- Built clinical benchmarking datasets for Reinforcement Learning tasks with the MIMIC-IV database.
- Implemented deep learning baseline of optimal dosing prediction for ICU medications (heparin and vancomycin).